Vision Pro

Presented By

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Apple leads the navigation of the changing tech seascape with Vision Pro

After facing the innovator's dilemma for the past decade, Apple is all set to launch its highly-anticipated headset "Vision Pro" scheduled for release in early 2024.

Vision Pro is priced at $3,499, targeting an elite segment of tech enthusiasts

The product stands out by placing a strong emphasis on Extended Reality (XR), setting it apart from the realm of virtual reality.

This innovative paradigm is labeled as "Spatial Computing," and the Vision Pro is officially termed a "Spatial Computer" with its own operating system VisionOS.

Apple faces an opportunity to shape the future of computing through the introduction of VisionPro but must tread the waters carefully.

Vision Pro's success depends on Social Concerns and Computational Advances

S

Concerns about addiction, health issues, and isolation (VR sickness impacts children. Calls for more research)

Consumer acceptance of wearables and business/social VR interactions.

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Advances in computing technology to power the growth of innovation in headsets

Advancement in headset form factors, optics, haptics, hand/body/eye tracking, AI and spatial computing

E

Favorable economic conditions and consumer discretionary spending required to support a premium priced product.

High price limits addressable market, lowering economies of scale.

E

Electronic waste impact from complex manufacturing and materials.

Carbon footprint from energy-intensive component manufacturing.

P

Potential political backlash against isolating effects of VR overuse.

Increased regulatory scrutiny on health effects, data privacy, content moderation. Age limits proposed.

Apple's aspirations for XR rely on Social & Health concerns and Computational Capabilities

Rapid advancements in computing technology

+

Tech's Double-Edged Sword: Immersive Advances Raise Concerns for Youth Wellbeing

In a tech-driven era, concerns about the social impact of immersive technologies have emerged. Despite advancements like body tracking and haptic feedback, studies reveal increased youth depression and addiction. Parents and medical professionals are urging further research into these effects, fueled by growing stories of addiction among the young and a call for stricter regulations.

Virtual Reality Breakthrough Sparks Global Frenzy

In the ever-evolving world of virtual reality, researchers cracked the code on reducing VR illness, turning it into a rare occurrence and triggering a global tech frenzy. As chip manufacturing increased, it bolstered the development of AR/VR, cranking up the immersion factor. Enthusiasts camped outside stores, waiting to snag the latest gear, and premium headset auctions turned into fierce battlegrounds.

ADVANCEMENTS IN COMPUTING TECHNOLOGY

Increased concerns around social & health impacts

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SOCIAL CONCERNS

+

Concerns around social & health impacts are addressed

VR Health Concerns Spark Global Outcry and Industry Setbacks

As of yesterday 1000s of children across the US have experienced health concerns from the increased use of Virtual Reality in their recreational lives. Indeed, parents all over the world are outraged and question the tech industry over their constant innovation. Meanwhile, the ongoing geopolitical tensions have impacted the manufacturing and availability of chips, putting a brake on the tech industry's rapid innovation.

VR triumphs worries but succumbs to stagnation

Contrary to fears of isolation and addiction, virtual reality (VR) is now celebrated for fostering social engagement and education. However, decreased access to chips has given rise to technical challenges. Including basic haptic feedback and less-than-realistic optics, temper the enthusiasm. Delays in AI-powered immersive experiences disappoint VR supporters, highlighting the industry's need to balance optimism with addressing technological shortcomings

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Slow growth in computing technology and advancements

Govt. laws and regulations promotes humanoid robots

+

GOVERNMENT REGULATIONS

SUPPLY CHAIN CONCERNS

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Govt. laws and regulations ban humanoid robots

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Disrupted supply of actuators

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Seamless Manufacturing and Supply of Actuators

AI loses its backbone as the world awaits the evolution

While government initiatives increased, public acceptance of AI and robots, supply chain disruptions have put a halt to this once-promising evolution, and geopolitical tech disputes are restricting access to essential components that fuel innovation.

Robotic Innovations: A Sustainable Future

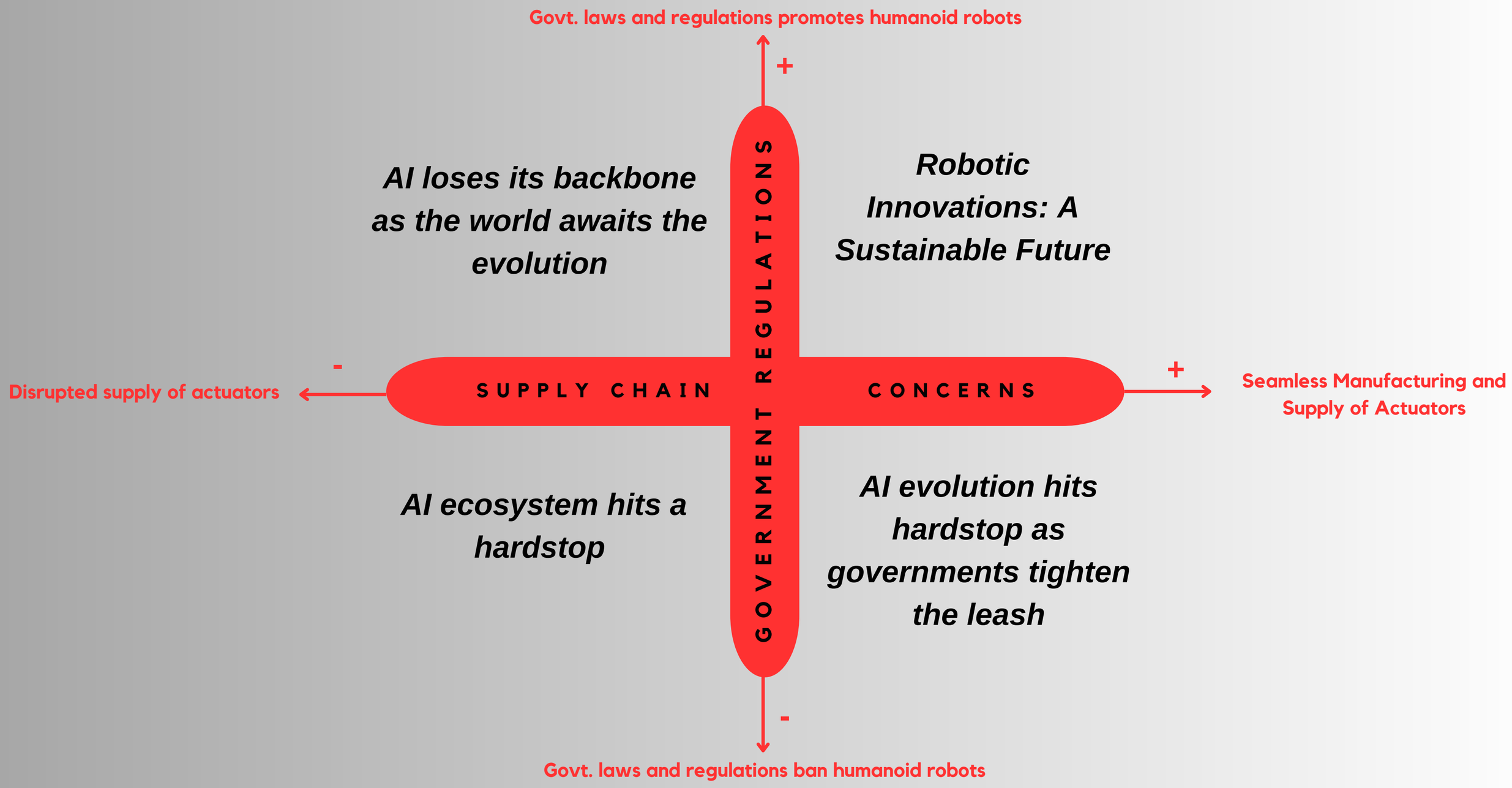
Humanoid robots are taking over routine tasks, heralding an exciting new era of possibilities marked by the swift integration of AI and robotics across industries and regulations that enable them. The prospects presented by this revolution in robotic technology are endless.

AI ecosystem hits a hardstop

Despite their widespread use, robots and artificial intelligence are falling short of expectations, which has led international leaders to discuss limiting their unbridled growth. Meanwhile, supply chain disruptions erode the ecology supporting AI. The next big thing has abruptly ended.

AI evolution hits hardstop as governments tighten the leash

Realizing the risks of unrestrained technological growth, world governments are taking action to limit the swift and uncontrolled proliferation of artificial intelligence and robots, which poses a threat to jobs. Global demonstrations have compelled innovation to be regulated in order to protect human welfare.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Addressing social concerns and computing power dependency drive VisionPro's growth before the arena becomes Red Sea

D

DRIVERS

- Social concerns about addiction, health issues, and isolation
- Advances in computing technology to power the growth of innovation in headsets

O

OPPORTUNITIES

- Shape the AR/VR/XR industry
- Build an Apple XR ecosystem
- Reboot the Metaverse

T

THREATS

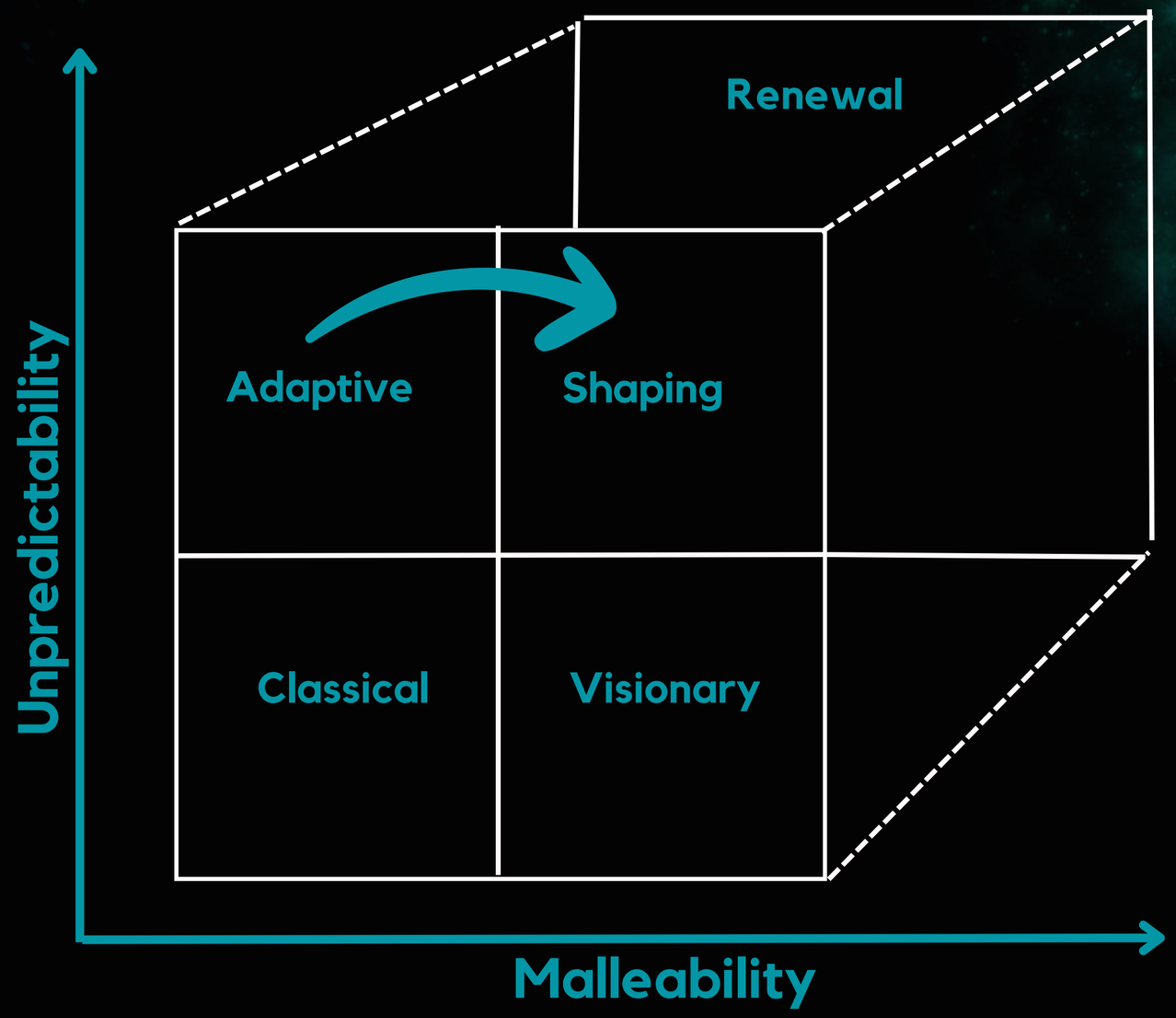
- Lack of computational power
- User dissatisfaction due to social and health concerns
- Competition from VR headsets

S

STRATEGIC RESPONSES

- Apple should focus on Market Penetration, Developer Partnerships, and Ethical Concerns
- Next, Apple should focus on Social Concerns, Vertical Integration, and Product Depth
- Apple should shape the future of Spatial Computing by focusing on Product Design, Enterprise Applications and the AI x Metaverse

Apple has the potential to move from an adaptive to shaping position as it delves into Spatial Computing



Adaptive Strategy

- Apple has been observing the VR/AR/XR space for a decade, waiting for the right moment to enter. This cautious approach aligns with adaptive strategy, where a company closely monitors industry developments and adapts its actions accordingly.

- The mention of Apple's partnership with Google on AI (Ajax) and its use of Vision Pro to integrate AI into its products suggests an adaptive response to technological advancements in the AI field.

Shaping Strategy

- Apple's attempt to shape the future of computing by entering the VR/AR/XR space with Vision Pro, VisionOS, and AI integration demonstrates a shaping strategy. The company is actively trying to influence the direction of the technology landscape and user behavior.

- Apple's emphasis on privacy and its potential to establish a new level of trust in immersive technologies aligns with the shaping strategy of setting industry standards and practices.

Apple faces pricing pressure from Sony/Pico, feature differentiation by Microsoft/XReal, and brand affinity of Meta in the race to define the AR/VR market

Incumbents

htc

Microsoft HoloLens

PICO

XREAL

Insurgents

NIANTIC

Magic Leap

Adjacents

Qualcomm

Tencent 腾讯

Brand Driven	
Distingusing Feature	XREAL, Microsoft HoloLens
Price	PICO

Vision Pro is aiming to deliver cutting-edge AR/VR through premium design and seamless ecosystem integration

Key Partners

Component suppliers (sensors, lenses, displays, batteries)

Foxconn, Pegatron for manufacturing and assembly

Carriers and retailers for distribution

Key Activities

Design and engineering of hardware and software

Supply chain mgmt

Marketing and brand management

Key Resources

Hardware engineering, design Software and AI engineering expertise

Value Propositions

Immersive experience

Advanced XR capabilities for productivity and creativity

Seamless integration with Apple ecosystem

Customer Relationship

Apple Support and Care

Personalized service and support

Channel

Apple Stores

Apple Website

Online Merchants

Customer Segments

Early tech adopters

Apple brand loyalists

Creative professionals and early adopters

Cost Structure

R&D, Production, Marketing, Retail Stors and Staffing

Revenue Stream

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Hardware Sales

Recurring subscriptions fees

Vision Pro is aiming to deliver cutting-edge AR/VR through premium design and seamless ecosystem integration

Key Partners

Suppliers of robotics components - motors, batteries, sensors, actuators, AI chips, etc.

Manufacturing partners to assemble Optimus robots

Key Activities

Designing and engineering Optimus

Developing proprietary robotics hardware and AI software

Key Resources

Robotics and AI engineering talent

Patented technologies

Brand recognition

Value Propositions

Personal robots with useful capabilities

Easy to use assistant for households, businesses, healthcare, etc.

Platform for scale development of future robotics applications

Customer Relationship

Direct sales to consumers and businesses

Customizable robots

Channel

Online and retail stores

Direct business partnerships

Trade shows and events

Customer Segments

Tech enthusiasts and early adopters

Households looking for help with chores/companionship

Businesses seeking affordable automation

Education, healthcare, hospitality, etc.

Cost Structure

Research and development expenses, Cost of components and manufacturing, Marketing and sales costs, After-sales support and cloud services

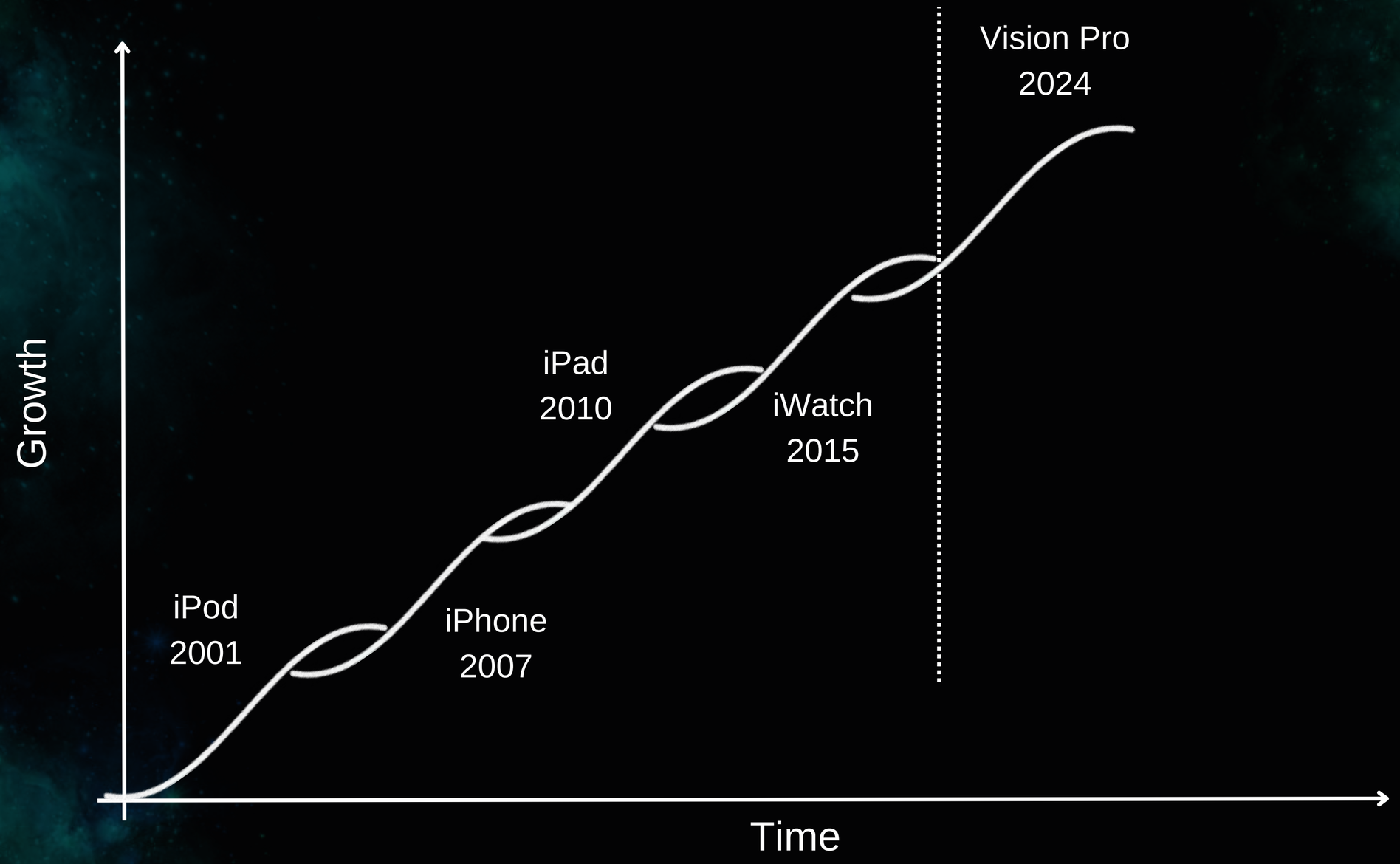
Revenue Stream

Robot sales to consumers and businesses

Subscription services for software, mapping, analytics etc.

Licensing and partnerships

Lifting the curve from groundbreaking technologies, Vision Pro is Apple's relentless pursuit of the next big thing



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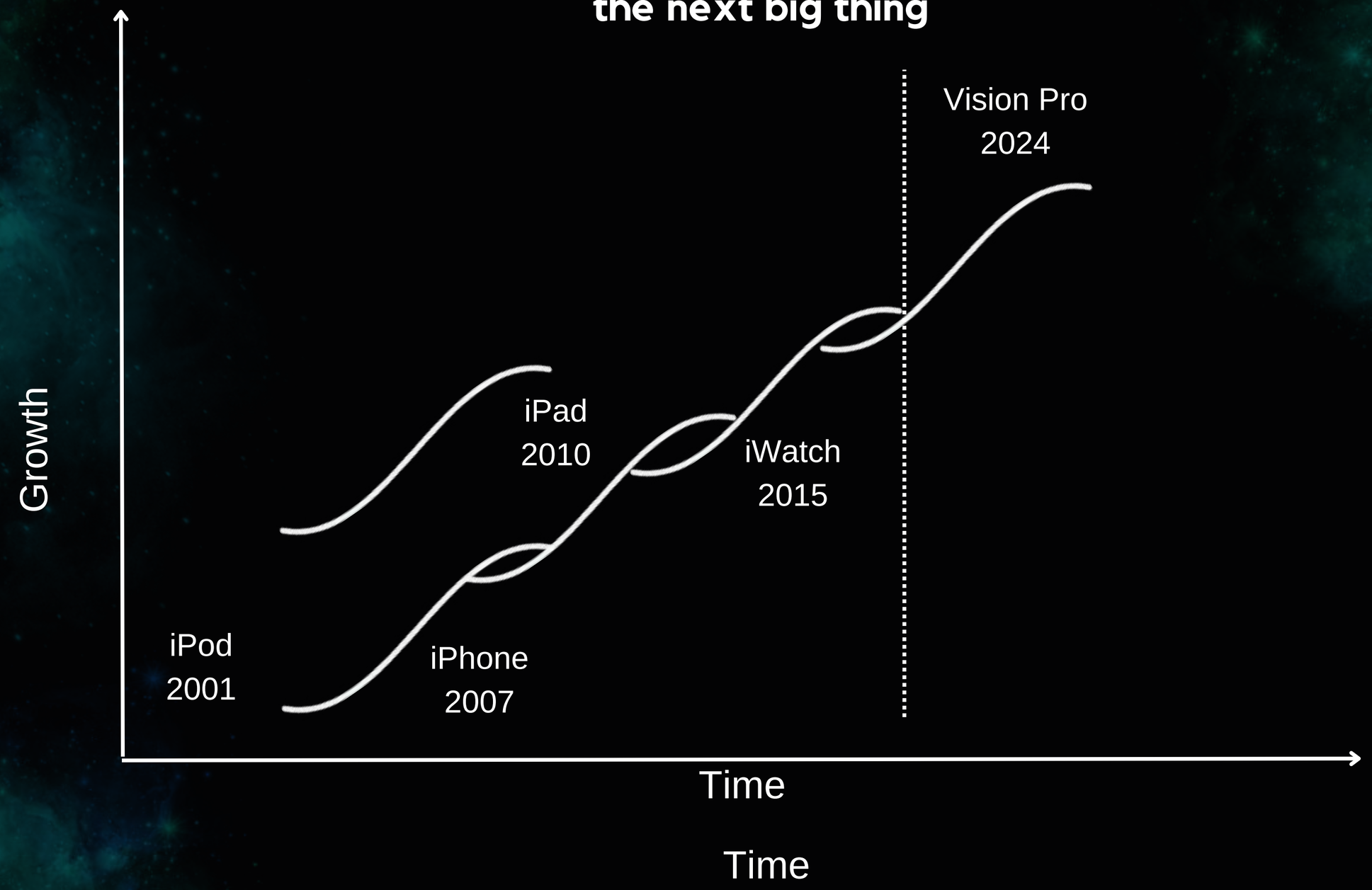
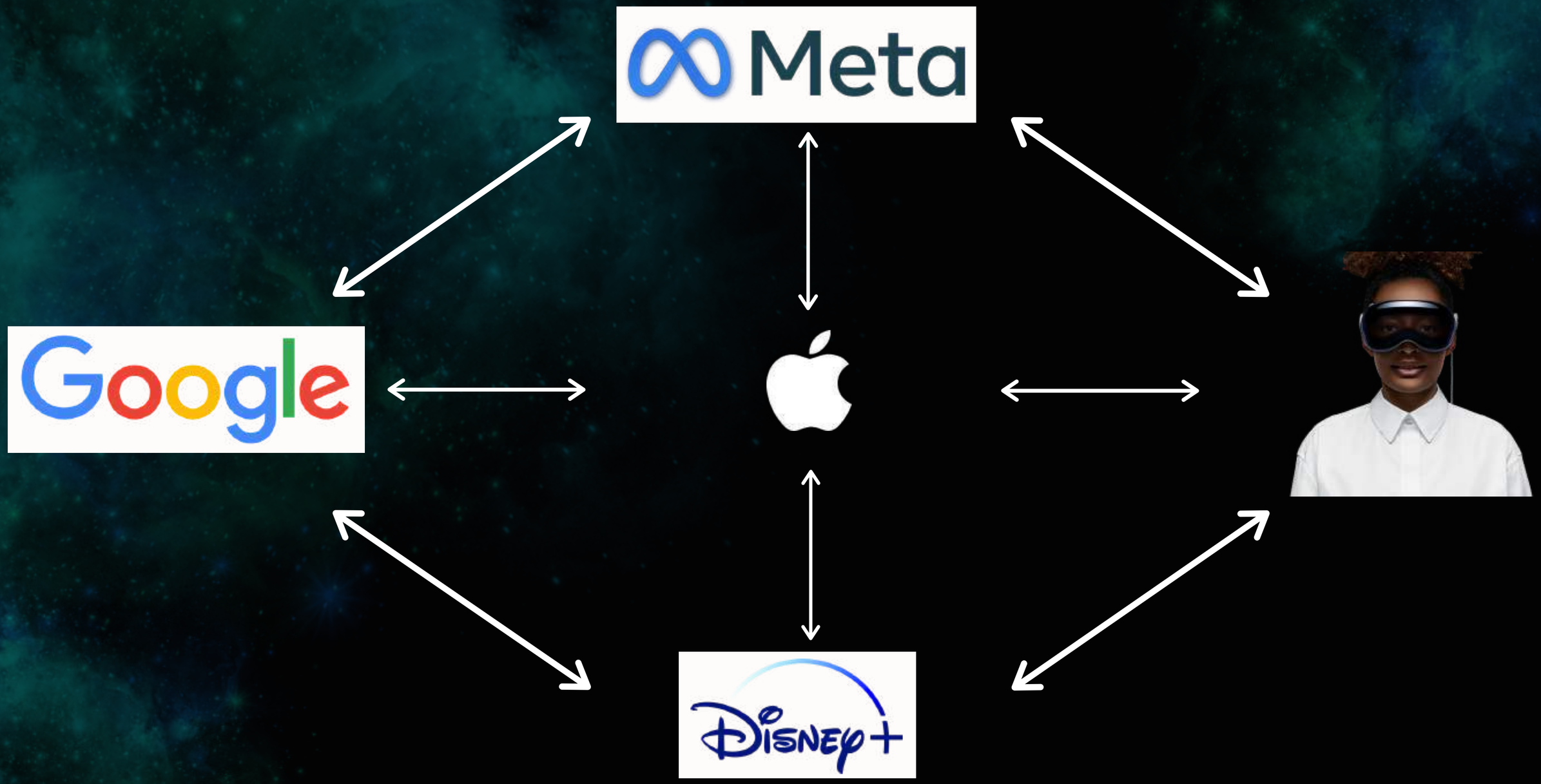
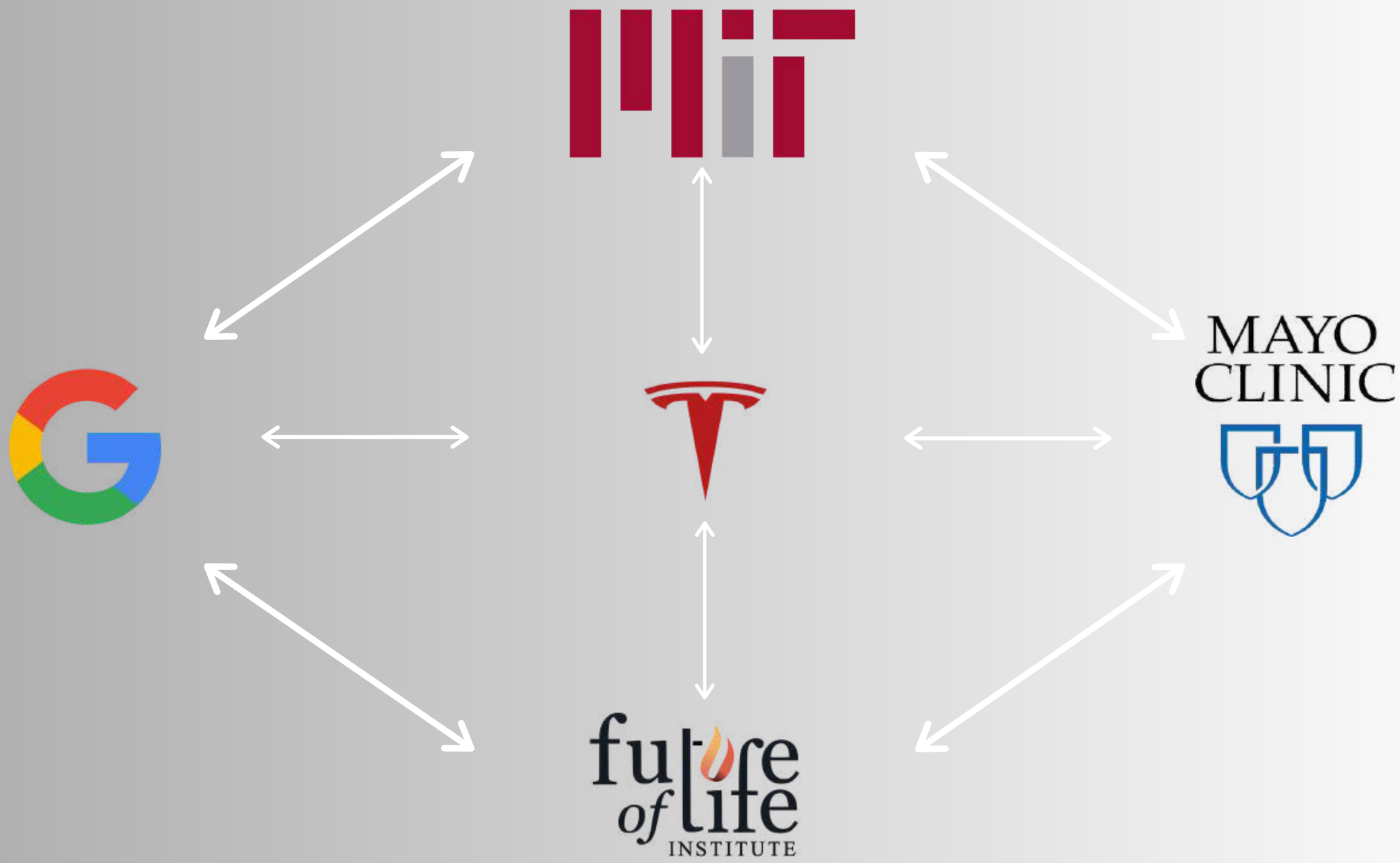


Apple should aim to position itself at the heart of Mixed Reality Ecosystem




MAYO
CLINIC
future
of life
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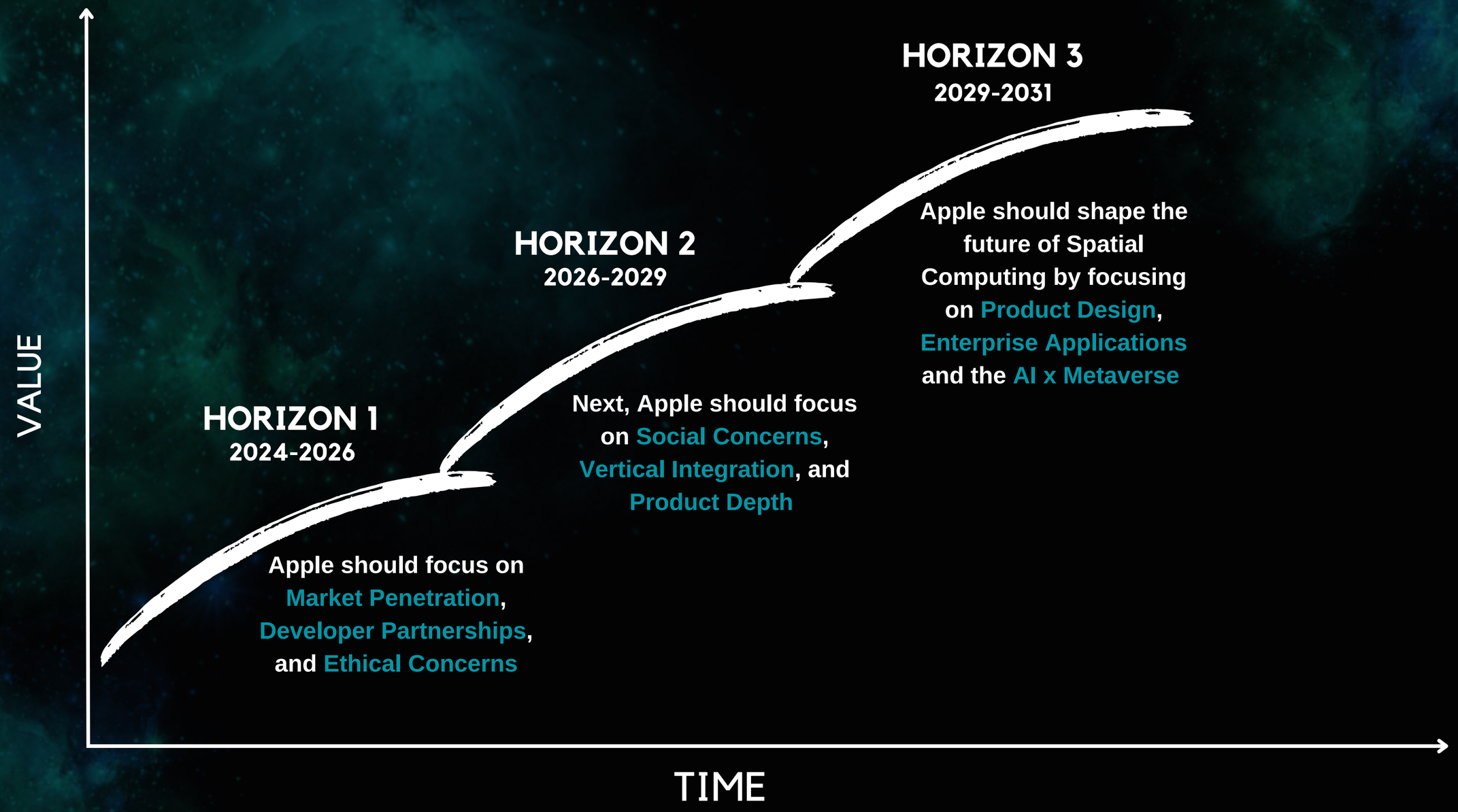
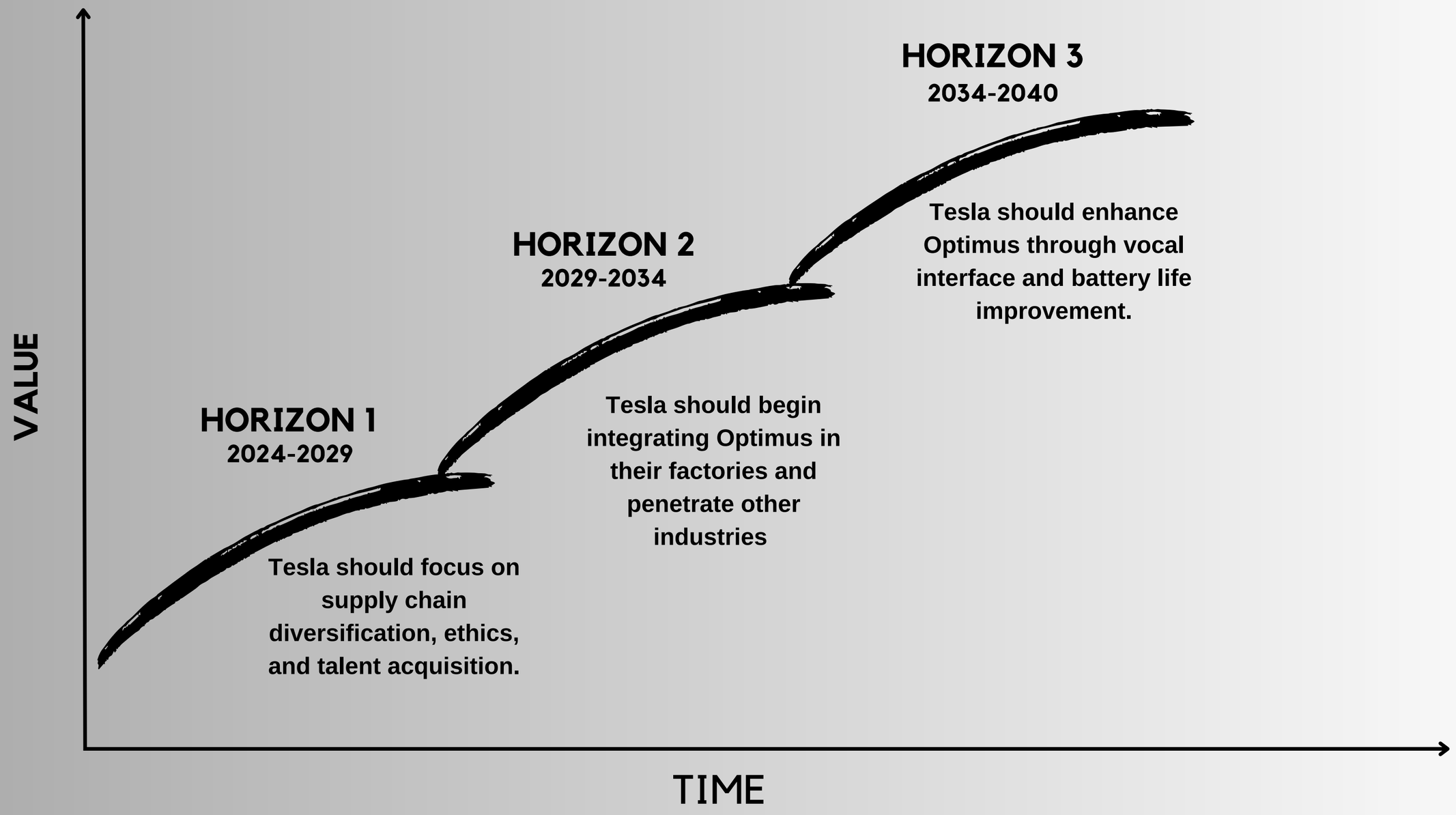
Building Strong Foundation, Improving Product and AI x Metaverse form the backbone of Apple's future




HORIZON 3
2034-2040
Tesla should enhance Optimus through vocal interface and battery life improvement.
HORIZON 2
2029-2034
Tesla should begin integrating Optimus in their factories and penetrate other industries
HORIZON 1
2024-2029
Tesla should focus on supply chain diversification, ethics, and talent acquisition.
VALUE
TIME

1 Apple should focus on Market Penetration, Developer Partnerships, and Ethical Concerns

How

Apple should focus on **aggressive marketing and expanding** the reach of the Apple Vision Pro in its target markets. Maintaining a steady **production line** to reach the projected sales.

Establish **developer partnerships** with companies, that provide a collaborative ecosystem to create content and provide software products designed for Vision Pro.

Anticipate and actively contribute to developing **ethical and regulatory frameworks** for AR/VR technologies and address any privacy concerns.

Why

The goal is to establish a **strong presence in the spatial computing arena** and drive early adoption into the new ecosystem.

To develop new **use cases** for all segments to be excited about Vision Pro and continue to support it in the future.

This is crucial to ensure **responsible use and protect user privacy** in the long term.

2 Next, Apple should focus on Social Concerns, Vertical Integration, and Product Depth

How

Dedicated R&D to address social and health concerns, providing feedback to develop the next-gen models. Some concerns are:

- **Addiction, isolation, motion sickness**
- Introduce **interpapillary distance (IPD)** or **diopter adjustment** in the lenses.

Expanding **vertical integration** by entering into joint ventures or acquisitions to ensure a steady supply of critical components, such as semiconductors and OLEDs.

Increase the product depth by introducing the **Pro and Pro Max models.**

Why

Focusing on user comfort and well-being, lowering the **risk of physical and social concerns**, hence boosting the entire user experience.

Reducing **reliance on external suppliers**, improving **control over production**, and **improving global supply chain challenges**, ultimately ensuring the availability of products and quality.

Presenting two versions of the product - **a flagship model and an affordable one** can help target a wider customer segment.

3 Apple should shape the future of Spatial Computing by focusing on Product Design, Enterprise Applications and the AI x Metaverse

How

Elevate **product design** by making it lightweight, fashionable, and more immersive.

Incorporate spatial computing to introduce **enterprise applications** for industry workers, surgeons, and other professionals, offering prompts to enhance efficiency in their tasks.

Commence extensive R&D efforts to create a **Metaverse environment** with Ajax's capabilities in the XR ecosystem.

Why

Improve user experience leading to AR/VR devices that merge seamlessly with fashion, **encouraging everyday use and public acceptance.**

Enabling transformational applications across several sectors illustrates the **technology's ability to improve various areas** of our daily lives.

Revolutionizing the arena with **new use cases** and becoming **shapers of the Metaverse**.

Executive Summary

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Vision Pro is aiming to deliver cutting-edge AR/VR through **premium design** and **seamless ecosystem integration**

Lifting the curve from groundbreaking technologies, Vision Pro is Apple's relentless pursuit to the next big thing

Apple positioning itself at the heart of Mixed reality ecosystem

Building Strong Foundation, **Improving Product** and **AI x Metaverse** form the backbone of Anthropic's future

Apple should focus on **Market Penetration**, **Developer Partnerships**, and **Ethical Concerns**

Next, Apple should focus on **Social Concerns**, **Vertical Integration**, and **Product Depth**

Apple should shape the future of Spatial Computing by focusing on **Product Design, Enterprise Applications and the AI x Metaverse**

LIMITLESS IMMERSION

APPLE VISION PRO

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